

Audio Amplifier Design Report

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1 Introduction

This project involves the design and construction of a four-stage audio amplifier capable of driving a $10\ \Omega$ speaker load with approximately 0.7 W of power. The design prioritizes minimal Total Harmonic Distortion (THD) and a flat frequency response within the audible range of 20 Hz to 20 kHz .

2 Differential Amplifier (Pre-Amplifier Stage)

2.1 Circuit Overview and Purpose

The Pre-amplifier uses a dual-input unbalanced output differential topology. Its primary role is to enhance weak signals (10 mV to 20 mV_{pp}) to line level while maintaining a high Signal-to-Noise Ratio (SNR) and suppressing common-mode noise.

2.2 Impedance Finding

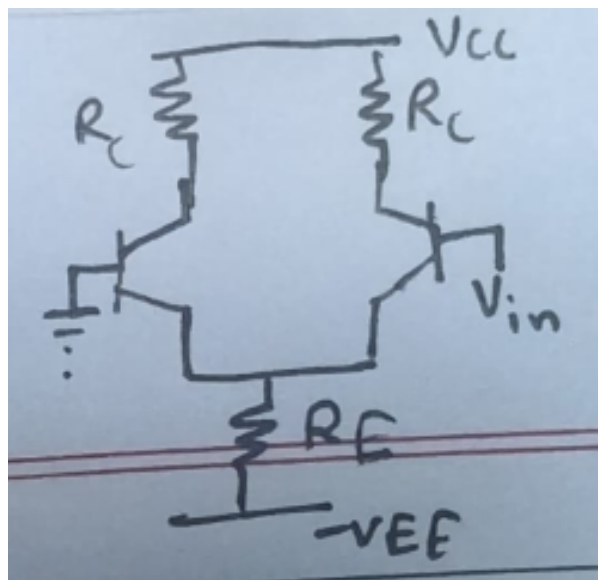


Figure 1: Circuit diagram

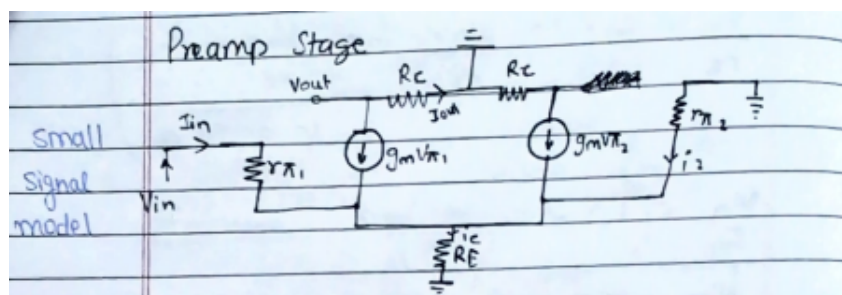


Figure 2: Small signal model of differential amplifier

The input impedance is defined as

$$Z_{in} = \frac{V_{in}}{I_{in}} \quad (1)$$

From the small signal model,

$$I_{in} = \frac{V_{in} - V_E}{r_{\pi 1}} \quad (1)$$

$$\frac{0 - V_E}{r_{\pi 2}} = -I_2 \quad (2)$$

Assuming

$$g_m v_{\pi 1} \gg I_1 \quad \text{and} \quad g_m v_{\pi 2} \gg I_2$$

Then,

$$V_E = g_m (v_{\pi 1} + v_{\pi 2}) R_E \quad (2)$$

Since

$$v_{\pi} = i r_{\pi}$$

we obtain

$$V_E = g_m R_E (I_{in} + I_2) r_{\pi} \quad (3)$$

From equation (1),

$$I_{in} = \frac{V_{in}}{r_{\pi}} - g_m R_E (I_{in} + I_2) \quad (4)$$

From equation (2),

$$I_2 = -\frac{g_m (I_{in} + I_2) R_E}{R_E} \quad (5)$$

Solving for I_2 ,

$$I_2 = -\frac{g_m I_{in} R_E}{1 + g_m R_E} \quad (6)$$

Substituting the value of I_2 ,

$$\frac{V_{in}}{r_{\pi}} = I_{in} \left(1 + g_m I_{in} R_E - \frac{g_m I_{in} R_E}{1 + g_m I_{in} R_E} \right) \quad (7)$$

Let

$$\alpha = g_m I_{in} R_E \quad \text{and} \quad \alpha \ll 1$$

Then,

$$\frac{V_{in}}{I_{in}} = r_{\pi} \left(1 + \alpha - \frac{\alpha}{1 + \alpha} \right) \quad (8)$$

For $\alpha \ll 1$,

$$\frac{V_{in}}{I_{in}} \approx r_{\pi} \quad (9)$$

Hence,

$$Z_{in} = r_{\pi}$$

which is the **input resistance**.

Output Resistance

$$R_{out} = \frac{V_{out}}{I_{out}}$$

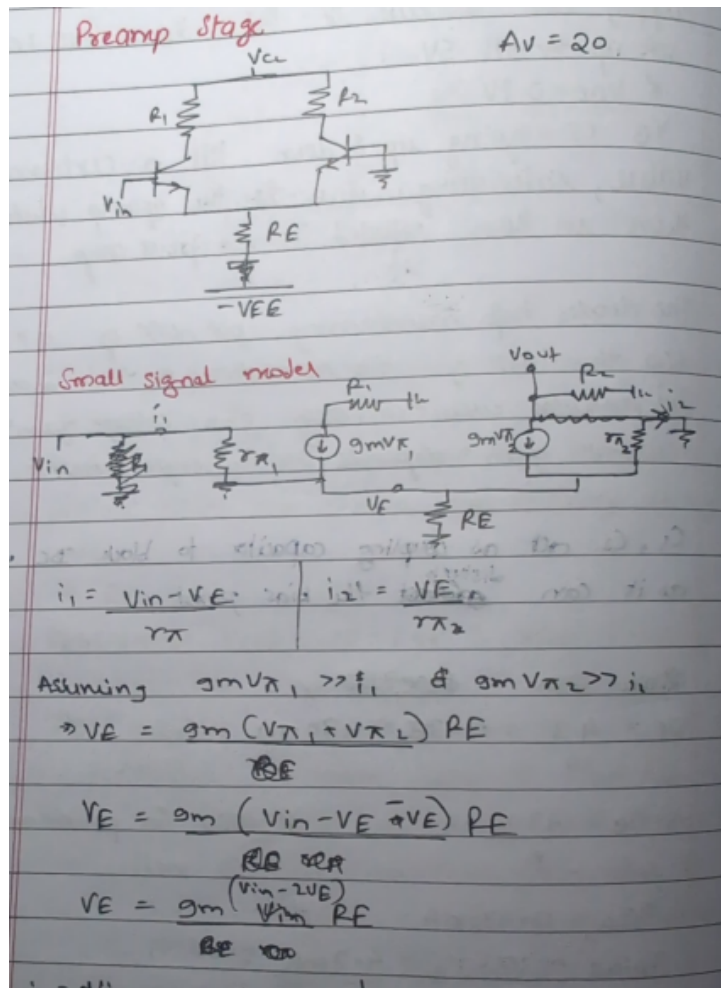
Since

$$V_{out} = I_{out} R_C$$

we obtain

$$R_{out} = R_C$$

2.3 Gain Derivation



$$\begin{aligned}
 V_E(1 + 2g_m R_E) &= g_m R_E V_{in} \\
 V_E &= \frac{g_m R_E V_{in}}{1 + 2g_m R_E} = -V_{\pi_2} \\
 \Rightarrow V_{out} &= +R_L = g_m \cdot \frac{g_m R_E V_{in}}{1 + 2g_m R_E} \approx 2g_m R_E \\
 \frac{V_{out}}{V_{in}} &= \frac{R_L g_m}{2} \\
 20 &= \frac{g_m(R_L)}{2} \quad R_L = 2.4 \text{ k}\Omega \\
 g_m &= \frac{20 \times 2}{2.4 \text{ k}} \Rightarrow I_C = \frac{26 \times 20 \times 10^{-6}}{2.4} \\
 I_C &\approx 0.433 \text{ mA} \\
 I_E &= 0.866 \text{ mA} = 2I_C \\
 g_m &= \frac{0.433}{26} \approx 0.016 \\
 \cancel{V_E} &= \frac{V_E \approx +V_{in}}{2} \\
 \text{Large Signal model} \\
 V_E &= 0.7 \text{ V, as } V_{BE} = 0.7 \text{ V} \\
 \therefore V_E - 0.7 + 5 &= 0.866 \text{ mA} \times R_E \\
 \Rightarrow \frac{-4.3 \text{ k}\Omega}{0.866} &= R_E \approx 5 \text{ k}\Omega \\
 \text{as mentioned above, } I_C \uparrow \Rightarrow I_E \uparrow \Rightarrow R_E \downarrow, \text{ about } 3.3 \text{ k}\Omega \\
 \text{taken for simula.}
 \end{aligned}$$

2.4 CMRR and Impedance Analysis

The Common Mode Rejection Ratio (CMRR) is critical for noise immunity:

$$\text{CMRR}_{abs} = g_m R_E \quad (10)$$

With $R_E = 3.3 \text{ k}\Omega$, the theoretical CMRR is 38.4 dB. The input impedance is high ($R_{in} = \frac{2\beta}{g_m}$), and output impedance is approximately R_C .

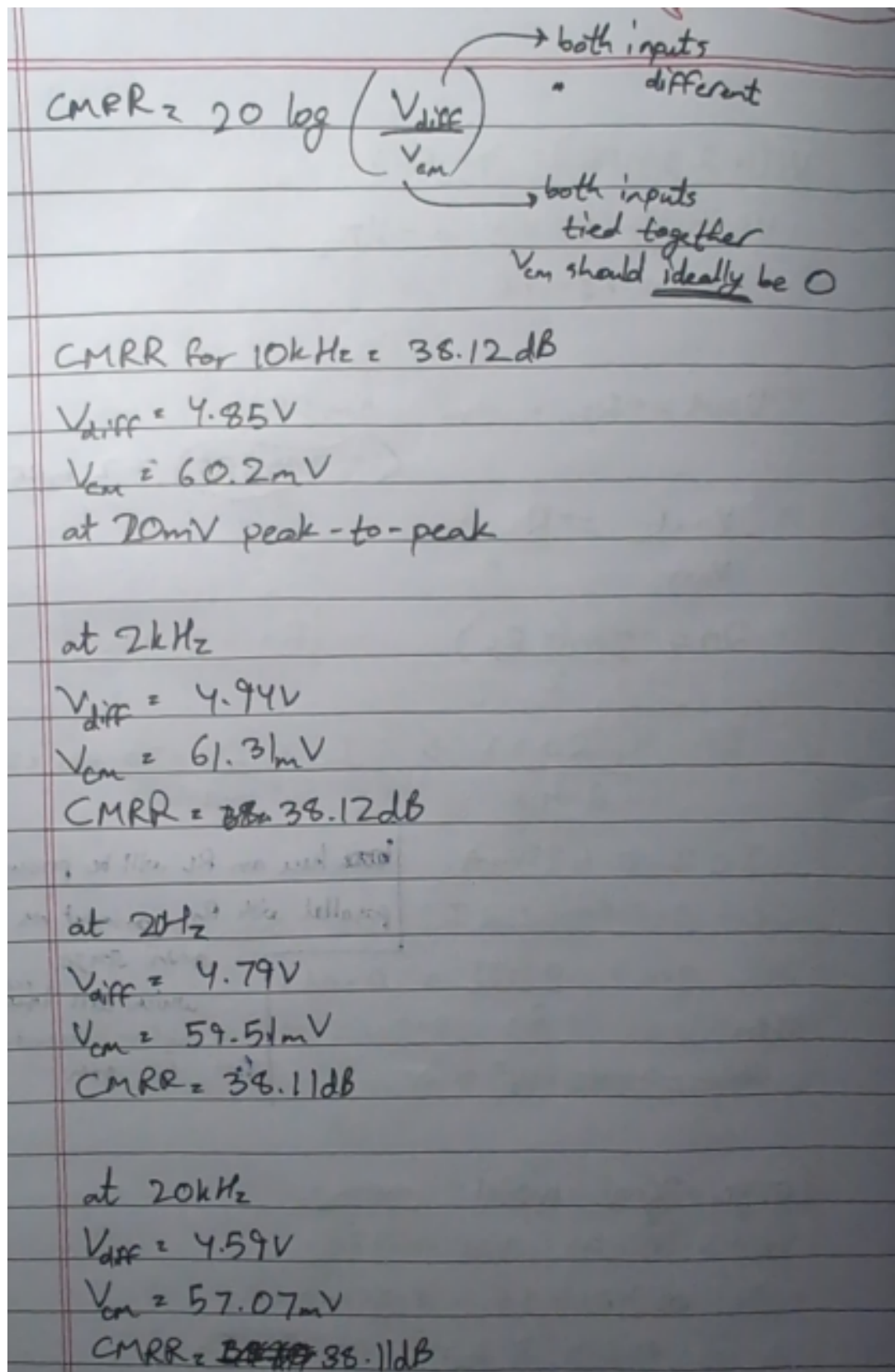


Figure 3: CMRR values for different frequencies

2.5 Simulation Validation

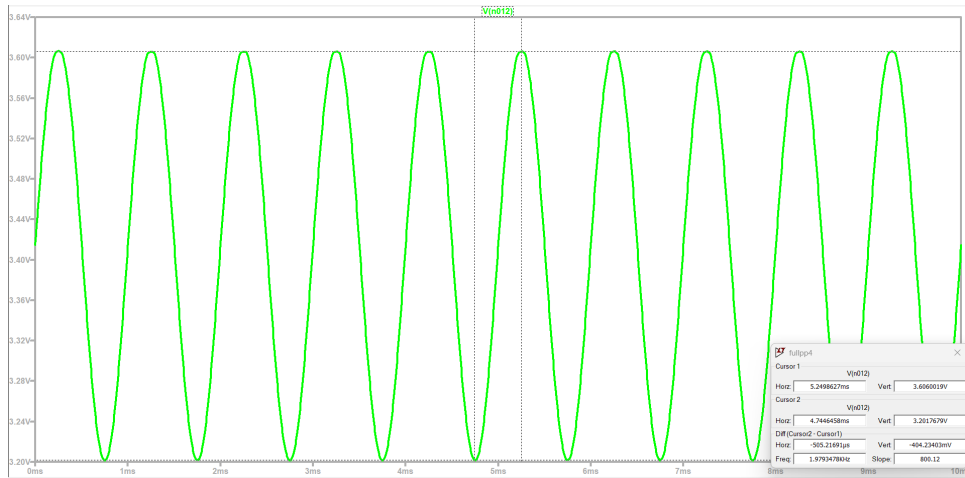


Figure 4: LTspice transient waveform — Differential Amplifier stage.

2.6 Measured LTspice Parameters — Pre-Amplifier

Table 1: LTspice Measured Parameters for Differential Stage

Parameter	Simulated Value
Collector Current (I_{C1}, I_{C2})	660 μ A
Quiescent Output Voltage ($V_{out,DC}$)	3.41 V
Differential Gain (A_d)	24.2
Common Mode Gain (A_c)	60.2
CMRR (Simulated)	38.12 dB

2.7 Hardware Implementation

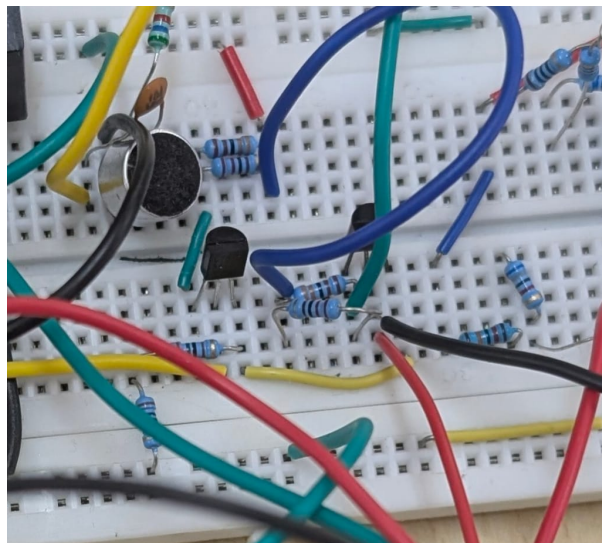


Figure 5: Hardware circuit of differential amplifier

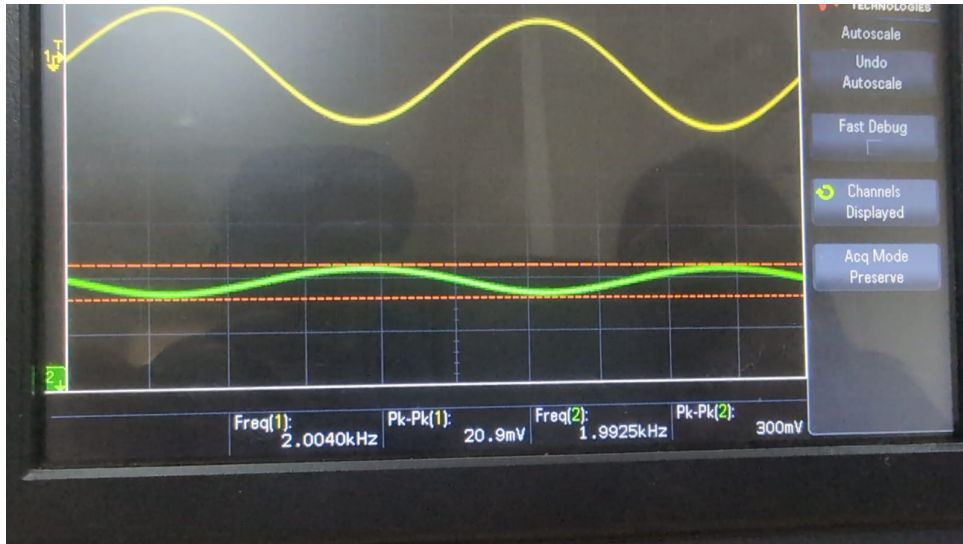


Figure 6: Waveform

3 Common-Emitter (CE) Gain Stage

3.1 Purpose and Biasing

The gain stage boosts the signal to a level suitable for the power amplifier. It utilizes voltage divider biasing to ensure the transistor remains in the active region throughout the signal cycle.

3.2 Impedence finding

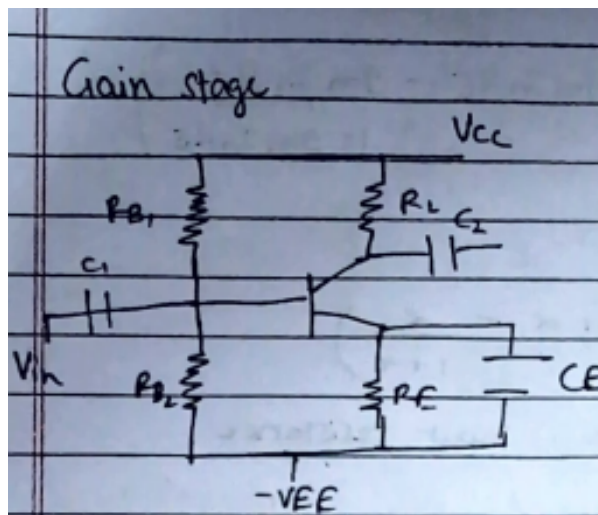


Figure 7: Circuit diagram of Gain stage

3.2.1 Output Resistance

To find the output resistance, set the input signal to zero.

$$V_{in} = 0$$

Hence,

$$g_m v_\pi = 0$$

Therefore the controlled current source becomes zero and the output current flows only through R_L .

$$I_{out} = \frac{V_{out}}{R_L}$$

Thus,

$$R_{out} = \frac{V_{out}}{I_{out}} = R_L$$

For the given circuit,

$$R_{out} = R_L$$

Small Signal Model

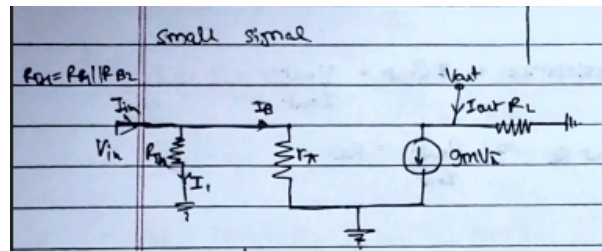


Figure 8: Small signal model

$$R_{Th} = R_{B1} \parallel R_{B2}$$

Current through R_{Th} :

$$I_1 = \frac{V_{in}}{R_{Th}}$$

Base current through r_π :

$$I_2 = \frac{V_{in}}{r_\pi}$$

Total current entering the node is

$$I = I_1 + I_2$$

$$I = \frac{V_{in}}{R_{Th}} + \frac{V_{in}}{r_\pi}$$

Input Resistance of Gain Stage

The input resistance is

$$R_{in} = R_{B1} \parallel R_{B2} \parallel r_{\pi}$$

Since

$$r_{\pi} \gg R_{B1}, R_{B2}$$

we approximate

$$R_{in} \approx R_{B1} \parallel R_{B2}$$

$$R_{B1} \parallel R_{B2} = \frac{R_{B1}R_{B2}}{R_{B1} + R_{B2}}$$

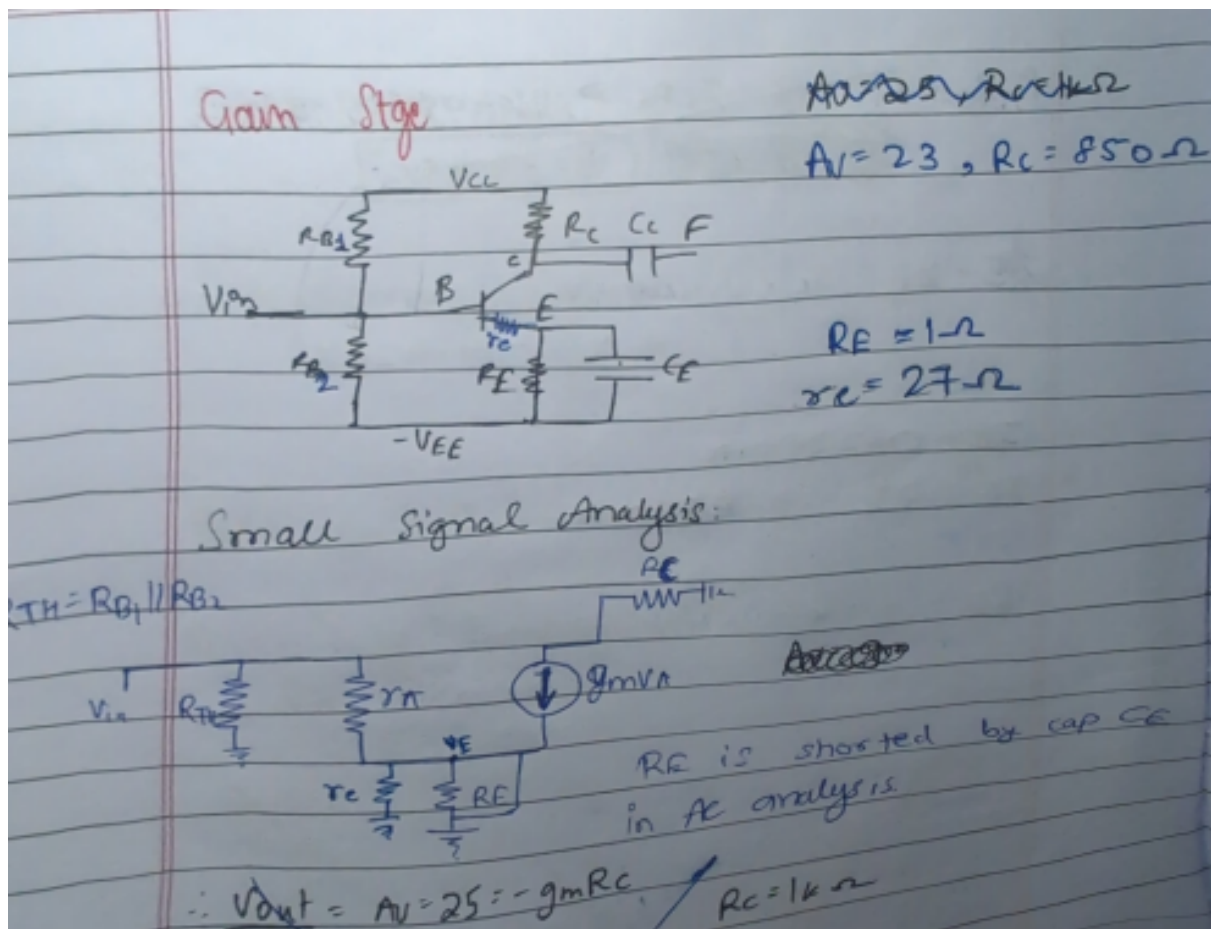
From the design,

$$\frac{R_{B1}R_{B2}}{R_{B1} + R_{B2}} \approx 10R_E \text{ (from preamp condition)}$$

Given

$$\frac{R_{B1}R_{B2}}{R_{B1} + R_{B2}} \approx 50 \text{ k}\Omega$$

3.3 Small-Signal Gain Derivation



$V_B = V_E + 0.7V$ assuming forward active Biasing

~~$V_{BE} = V_B - V_E = 0.7V$~~ ~~$R_B = R_1 \parallel R_2$~~ =

Bias current atleast 10 times the I_B [Base current] for stable biasing.

$$I_B = \frac{I_C}{\beta} \quad | \quad I_{Bias} = 10 \times I_B$$

R_E here is r_e in circuit

$$\frac{V_{in} - V_E}{r_x} + g_m V_E = \frac{V_E}{R_E} \quad | \quad V_{out} = -g_m V_E R_C$$

$$V_E = V_{in} - V_E$$

$$\Rightarrow V_{in} - V_E \left[g_m + \frac{1}{r_x} \right] = \frac{V_E}{R_E} \quad | \quad V_{out} = -g_m R_C (V_{in} - V_E)$$

$$\Rightarrow V_E = V_{in} + \frac{V_{out}}{g_m R_C}$$

$$\Rightarrow \frac{-V_{out}}{g_m R_C r_x} + g_m \left(\frac{-V_{out}}{g_m R_C} \right) = \frac{V_{in}}{R_E} + \frac{V_{out}}{g_m R_C R_E}$$

$$\Rightarrow \frac{-V_{out}}{V_{in}} = \frac{1}{\frac{1}{g_m R_C r_x} + \frac{1}{R_C} + \frac{1}{g_m R_C R_E}}$$

$$= \frac{R_C}{R_E \left(\frac{1}{g_m R_C r_x} + \frac{1}{g_m R_C} + 1 \right)} \Rightarrow \frac{-V_{out}}{V_{in}} = \frac{R_C}{R_E \left(1 + \frac{1}{g_m R_E} \right)}$$

$$\Rightarrow \boxed{A_v = \frac{-R_C R_E}{R_E + 1/g_m}} \Rightarrow A_v = \frac{-R_C}{R_E + 1/g_m}$$

Large Signal

setting $V_{base} = -4.16V$

setting $V_E = -4.16 - 0.7 = -4.86V$

V_E should be as low as possible to get max voltage swing at V_C

$$I_E = \frac{0.14}{28} = 5.3mA \text{ [matching simulation]}$$

Approx, $I_E = I_C = 5.3mA$

$$A_v = 23$$

$$V_{out} = 5.3 \times R_C$$

$$\frac{V_{out}}{V_{in}} = 23$$

V_{out} should $\Rightarrow 4.6 = 5.3mA \times R_C$

max swing of $\Rightarrow R_C = \frac{4.6}{5.3}$

$9.2V \Rightarrow 4.6V$

is the amp.

$$\therefore R_C = 867\Omega$$

$$\begin{aligned} V_{out} &= 23 \times 200mV \\ &= 4600mV \\ &= 4.6V \end{aligned}$$

$$V_{base} = -4.16V$$

$$\beta = 300 \Rightarrow I_b = \frac{5.3mA}{300}$$

Approximating Base biasing as voltage divider

$$I_b = 17.66\mu A$$

base current

$$V_B = \frac{R_{b2} \times (V_{CC} - V_{EE}) + V_{EE}}{R_{b1} + R_{b2}}$$

$$-4.16 = \frac{R_{b2}(10) - 5}{R_{b1} + R_{b2}} \Rightarrow 0.84 = \frac{10R_{b2}}{R_{b2} + R_{b1}}$$

$$\Rightarrow R_{b1} \cdot 0.84 = 9.16 R_{b2}$$

$$\frac{R_{b1}}{R_{b2}} = \frac{9.16}{0.84} = 11.45$$

$$R_{b2} = 17.6k\Omega \Rightarrow R_{b1} = 198k\Omega$$

3.4 Integration and Verification with Stage 1

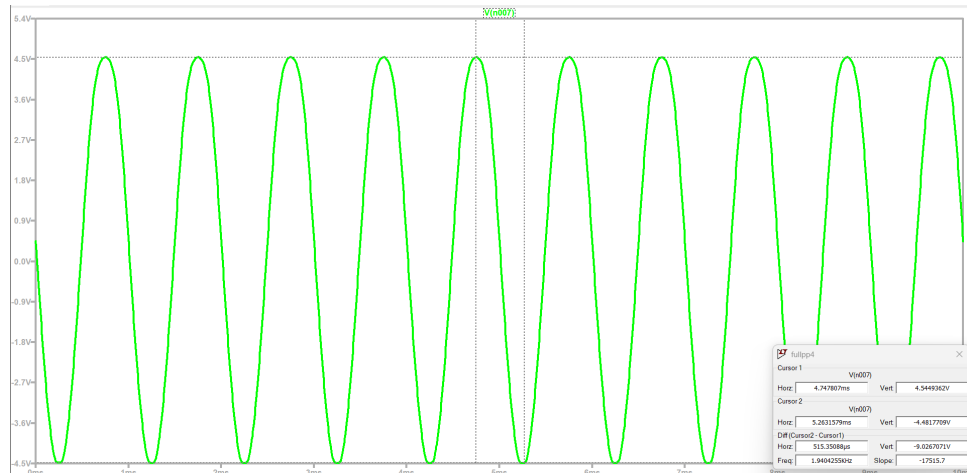


Figure 9: LTspice waveform — Stages 1 & 2 cascaded output.

3.5 Measured LTspice Parameters — CE Stage

Table 2: LTspice Measured Parameters for Gain Stage

Parameter	Simulated Value
Base Voltage (V_B)	-4.16 V
Collector-Emitter Voltage (V_{CEQ})	5.31 V
Collector Quiescent Current (I_{CQ})	5.35 mA
Voltage Gain (A_v)	26 dB
Input Impedance (Z_{in})	15.63 k Ω

3.6 Hardware Implementation

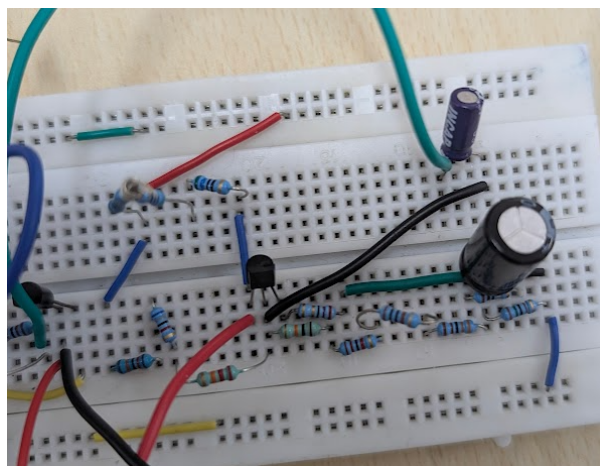


Figure 10: Hardware circuit for gain stage



Figure 11: Waveform

4 Filter Stage (20 Hz–20 kHz)

4.1 Active Band-Pass Implementation

An active filter is used to maintain signal strength while eliminating low-frequency hum and high-frequency interference. A second-order filter was created by cascading two first-order stages to achieve a steeper roll-off.

Input and Output Impedance

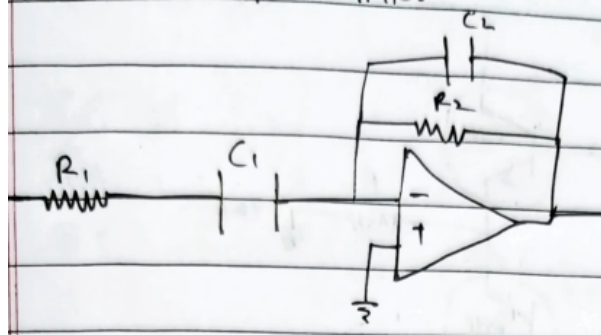


Figure 12: Circuit diagram for the active bandpass filter

The input resistance of the bandpass filter is

$$R_{in} = R_1 + \frac{1}{j\omega C_1}$$

Since R_1 and C_1 form a high-pass filter, the cutoff frequency is

$$f_c = \frac{1}{2\pi R_1 C_1}$$

For operating frequencies much higher than the cutoff frequency,

$$\frac{1}{j\omega C_1} \ll R_1$$

Hence,

$$R_{in} \approx R_1$$

To find the output resistance,

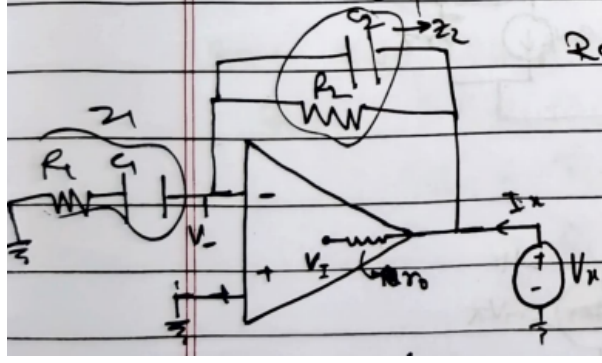


Figure 13: Circuit diagram to find output impedance

$$R_{out} = \frac{V_x}{I_x}$$

Assume the op-amp has a finite open-loop gain

$$A_v \approx 10^5 - 10^6$$

The output resistance of the op-amp itself is very small,

$$r_o \approx 100 \Omega$$

Let

$$Z_1 = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

The node voltage at the inverting input is

$$V_- = V_x \frac{Z_1}{Z_1 + Z_2} \quad (11)$$

Since the non-inverting terminal is grounded,

$$V_+ = 0$$

The internal op-amp voltage is

$$V_o = A_v(V_+ - V_-)$$

$$V_o = -A_v V_x$$

The test current becomes

$$I_x = \frac{V_x}{Z_1 + Z_2} + \frac{V_x - V_o}{r_o} \quad (12)$$

Substituting V_o ,

$$I_x = \frac{V_x}{Z_1 + Z_2} + \frac{V_x(1 + A_v)}{r_o} \quad (13)$$

Thus the output resistance becomes

$$R_{out} = \frac{V_x}{I_x}$$

$$R_{out} \approx \frac{r_o}{1 + A_v}$$

Since

$$A_v \gg 1$$

$$R_{out} \approx \frac{r_o}{A_v}$$

For typical values

$$r_o \approx 100 \, \Omega, \quad A_v \approx 10^5$$

$$R_{out} \approx 10^{-3} \, \Omega$$

Thus the output resistance of the active bandpass filter is extremely small.

4.2 Transfer Function

Active Filter BandPass

high pass filter

low pass filter

C_1 blocks low frequencies, $Z_C = \frac{1}{2\pi f C_1}$

low f $\Rightarrow$ high Impedance $\Rightarrow$ acts as open circuit for low freq.

$$f_c = \frac{1}{2\pi R_1 C_1}$$

At high frequencies C_2 acts as short circuit

$\Rightarrow$ shorts the R_2

$\therefore$ Z_{feedback} of opamp ≈ 0

$\Rightarrow$ gain drops to 0.

$$\text{gain} = \frac{Z_{\text{feedback}}}{Z_{\text{in}}}$$

$Z_{\text{in}} = R_1 + \frac{1}{j\omega C_1}$ | cutoff frequency is decided so -3dB gain is achieved for it

$\Rightarrow$ this happens when capacitive reactance is equal to resistance.

$$\frac{1}{j\omega C_1} = R_1 \Rightarrow f_c = \frac{1}{2\pi C_1 R_1}$$

Similarly, $Z_f = R_2 \parallel \frac{1}{j\omega C_2}$

cut-off occurs at gain = -3dB $\rightarrow R_2 = \frac{1}{j\omega C_2}$

$$\therefore f_H = \frac{1}{2\pi R_2 C_2}$$

For the middle band, i.e. the freq b/w f_L & f_H the capacitors provide certain impedance & ~~lets it pass with~~.

for that this scenario, Gain due to opamp

$$= -\frac{Z_f}{Z_{in}}$$

$$= -R_2 \parallel \frac{1}{j\omega C_2}$$

$$R_1 + \frac{1}{j\omega C_1}$$

Assuming that C_1 is almost short circuited, i.e. Provides very low impedance.

& C_2 is open, i.e., provides very high impedance.

$\rightarrow$ we can approximate the gain as $-\frac{R_2}{R_1}$

C_1 is high cap value, C_2 is low cap value.

for the LPF must be so that 20kHz passes without significant attenuation.

∴ $R_1 = 20\text{ k}\Omega$, ~~$f_H = 20\text{ kHz}$~~ $f_H = 53\text{ kHz}$

∴ $C_1 = 150\text{ pF}$ by $f_H = \frac{1}{2\pi R_1 C_1}$ 20 kHz
This f_H is selected to have minimal attenuation for 20kHz

As for the HPF, 20Hz f_{req} should pass with minimal attenuation.

~~Attenuation~~ Attenuation in here cannot be stopped i.e. a sharp filter cannot be created so as to have ~~zero~~ no attenuation at such low frequency.

Since the gain of opamp = $-\frac{R_{\text{feedback}}}{R_{\text{in}}}$

~~$R_{\text{in}} = R_F = 20\text{ k}\Omega$~~ , $R_{\text{in}} = 20\text{ k}\Omega$

ideally, $R_F > R_{\text{in}}$ as the open loop resistance of opamp will include the output res. of the previous stage, but this approx approximation works.

∴ $R_F = 20\text{ k}\Omega$, ~~$R_{\text{in}} = 20\text{ k}\Omega$~~

Cap of 10 μ is chosen so that the attenuation of the ω lower frequency can be minimized while ~~to~~ blocking DC.

∴ $f_L = \frac{1}{2\pi \times 10\mu \times 20\text{ k}} = 0.79\text{ Hz}$

4.3 Bode Plot and Gain Analysis

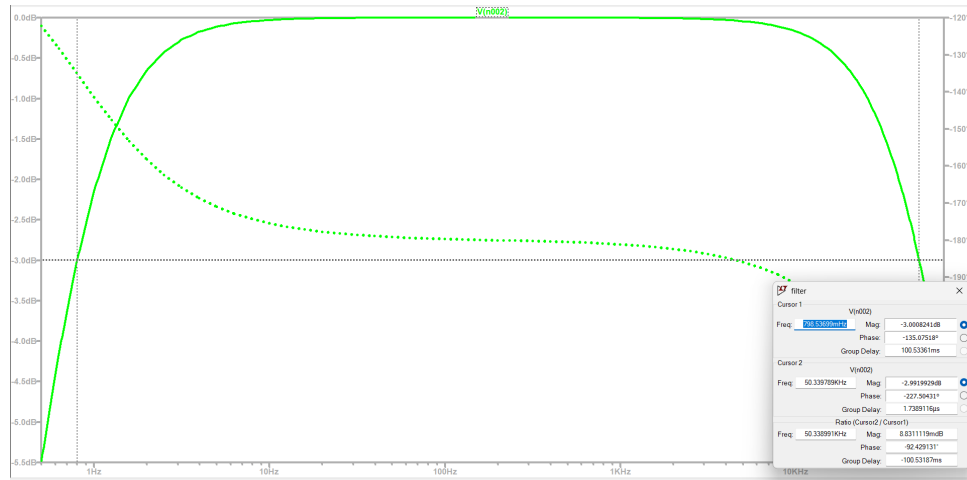


Figure 14: Bode plot of band-pass filter — 0 dB passband from 20 Hz to 20 kHz.

4.4 Measured LTspice Parameters — Filter Stage

Table 3: LTspice Frequency Response Data

Parameter	Simulated Value
Lower Cutoff Frequency (f_L)	0.8 Hz
Upper Cutoff Frequency (f_H)	50.3 kHz
Mid-band Gain	0 dB

4.5 Hardware Implementation

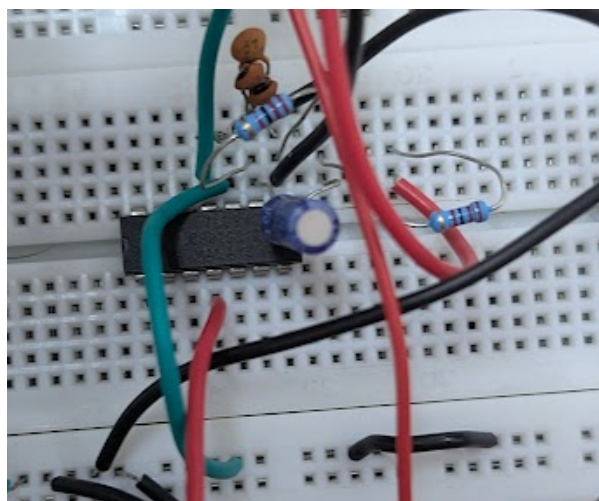


Figure 15: Active bandpass filter hardware

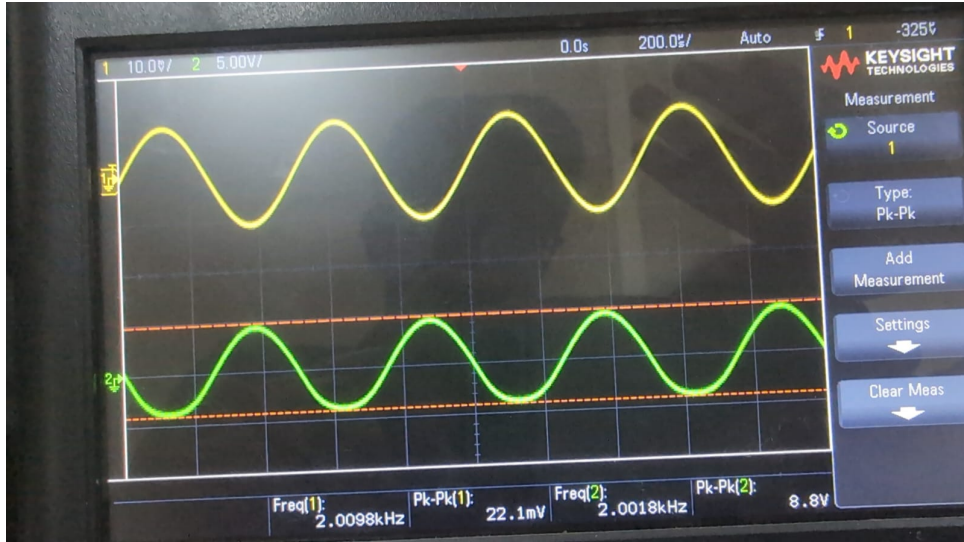


Figure 16: waveform

5 Power Amplifier Stage

5.1 Class AB Configuration

A Class AB push-pull configuration is utilized to drive the speaker load efficiently while eliminating crossover distortion. Diode-biasing is employed to keep the output transistors slightly "ON" even without an input signal.

5.2 Power and Efficiency

The stage is designed to drive a 10Ω load with a peak voltage swing approaching the $\pm 5\text{V}$ rails. It provides significant current gain (approx. 20) while maintaining a voltage gain of nearly unity.

5.3 Input and Output Impedance

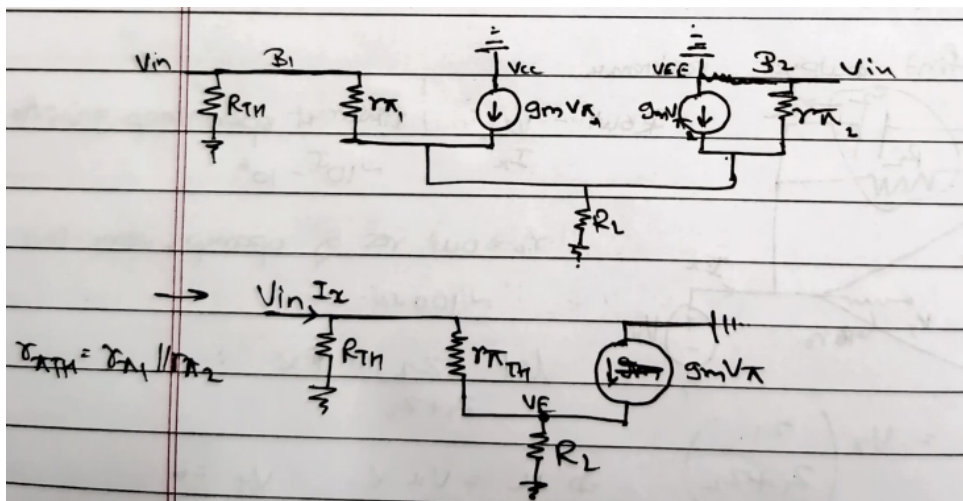


Figure 17: Small Signal Analysis of power amplifier

For small signal analysis,

$$V_{B1} = V_{B2}$$

Hence there is no potential difference across the bias resistances. Therefore the Thevenin resistance of the bias network is

$$R_{TH} = R_1 \parallel R_2$$

Small Signal Analysis

The current through the Thevenin resistance is

$$I_x = \frac{V_{in}}{R_{TH}}$$

The base-emitter voltage is

$$V_\pi = V_{in} - V_E$$

The emitter voltage is related to the emitter current through the load resistor R_L .

$$V_E = R_L I_E$$

The emitter current consists of the transistor current components

$$I_E = g_m V_\pi + \frac{V_\pi}{r_\pi}$$

Substituting V_π ,

$$V_E = R_L \left(g_m + \frac{1}{r_\pi} \right) (V_{in} - V_E)$$

Solving for V_E ,

$$V_E = \frac{R_L \left(g_m + \frac{1}{r_\pi} \right) V_{in}}{1 + R_L \left(g_m + \frac{1}{r_\pi} \right)}$$

Substituting this into the input current expression,

$$I_x = \frac{V_{in}}{R_{TH}} + \frac{V_{in}}{r_\pi} \left[\frac{1}{1 + R_L \left(g_m + \frac{1}{r_\pi} \right)} \right]$$

Thus,

$$I_x = \frac{V_{in}}{R_T}$$

where

$$R_T = R_{TH} \parallel (r_\pi + R_L (g_m r_\pi + 1))$$

Base Resistance due to Transistors

Looking into the emitter,

$$I_E = (\beta + 1)I_B$$

Thus the resistance seen at the base is

$$R_B = (\beta + 1)R_E$$

Including the emitter resistance r_e ,

$$R_B = (\beta + 1)(R_L + r_e)$$

Since only one transistor conducts at a time in the push-pull configuration,

$$R_B \approx (\beta + 1)(R_L + r_e)$$

Input Resistance

The input resistance of the power amplifier is

$$R_{in} = R_1 \parallel R_2 \parallel R_B$$

Substituting R_B ,

$$R_{in} = R_1 \parallel R_2 \parallel (\beta + 1)(R_L + r_e)$$

Output Resistance

$$R_{out} = R_L$$

5.4 Derivations

Power Amplifier

$\rightarrow$ Voltage gain $= 1$, maximize maximizing power gain
 $\Rightarrow$ increasing the current
 $\Rightarrow$ if Power Amp has high input res, the op-amp of previous stage can provide the necessary driving current.
 $\Rightarrow R_{B1} \& R_{B2} \uparrow$
~~V_{BE1} = 10 - 1.4V~~ $V_{RES} = 10 - 1.4V \rightarrow V_{RES} = 8.6V$
 $\therefore V_{RB1} = V_{RB2} = 4.3V$
 $R_{B1} = \frac{V_{RB1}}{I_{B1}}$
 $I_{B1} = I_{B2}$ Assume I_{B1} & find R_{B1}
 $I_{B1} = 4.2mA$ - NPN is better than PNP transistor
 $\Rightarrow$ better current gain due to higher mobility of e^- than holes
 $I_{B1} = 4.2mA$
 $\Rightarrow R_{B1} = \frac{4.3}{4.2mA} \approx 1k\Omega = R_{B2}$

clipping here is due to the V_{base} not being set up to till 5V.

$$\& V_{BE} = 0.7V.$$

V_B is going up & down till a certain value, saturating due to the opamp which is then in turn reflected in the power amp.

The diodes help maintaining pot diff of 0.7 to 0.7 b/w the BE of the transistors or 1.4V as a whole.

This particular circuit can only give Voltage gain = 1
current gain depends on the input current

C_1, C_2 act as coupling capacitor to block DC voltage as it can ^{disturb} ~~affect~~ the bias point

$$I_E = \frac{4.3}{10} = 0.43A = 430mA$$

$$\therefore I_B = \frac{430mA}{\beta}$$

$\beta = 1000$ for power amp
(avg val)

$$\Rightarrow I_B = 0.43mA$$

$$I_{Bias} \sim 10 \times I_B = \underline{4.3mA}$$

5.5 LTspice Simulation Analysis

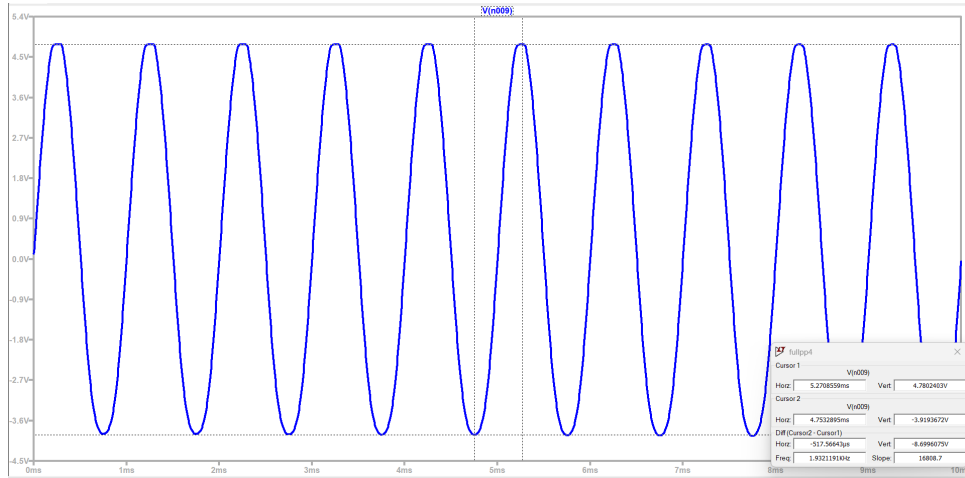


Figure 18: LTspice analysis of Power Stage — Output voltage swing and crossover check.

5.6 Measured LTspice Parameters — Power Stage

Table 4: LTspice Measured Parameters for Class AB Stage

Parameter	Simulated Value
Quiescent Current (I_Q)	-12.3 mA
Maximum Peak Voltage Swing	9.2 V
Voltage Gain (A_v)	≈ 1
Output Power (P_{out})	1.05 W

5.7 Hardware Implementation

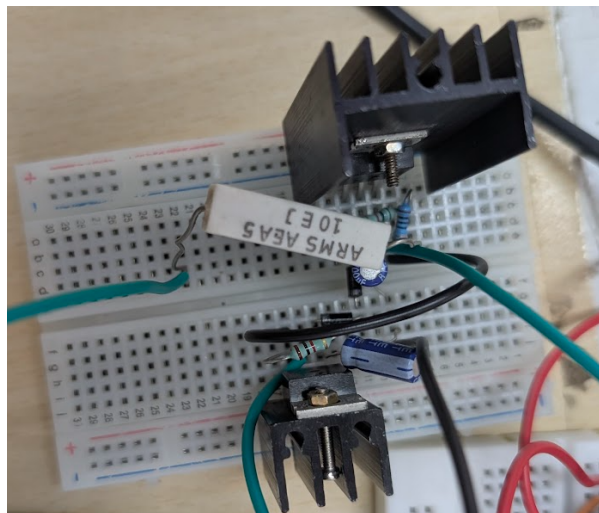


Figure 19: Power amplifier hardware

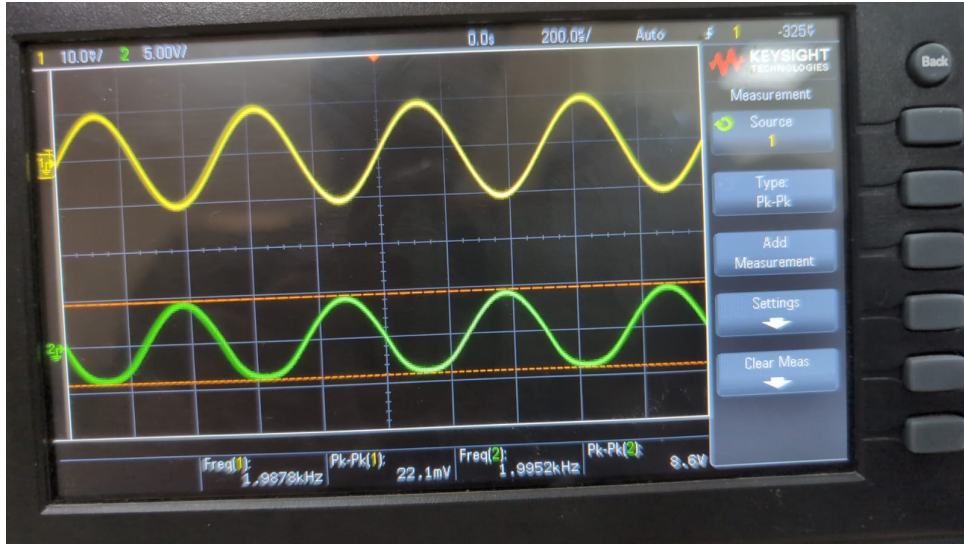


Figure 20: Waveform

6 Overall Performance and Analysis

6.1 Final Gain Verification

6.1.1 AC Analysis

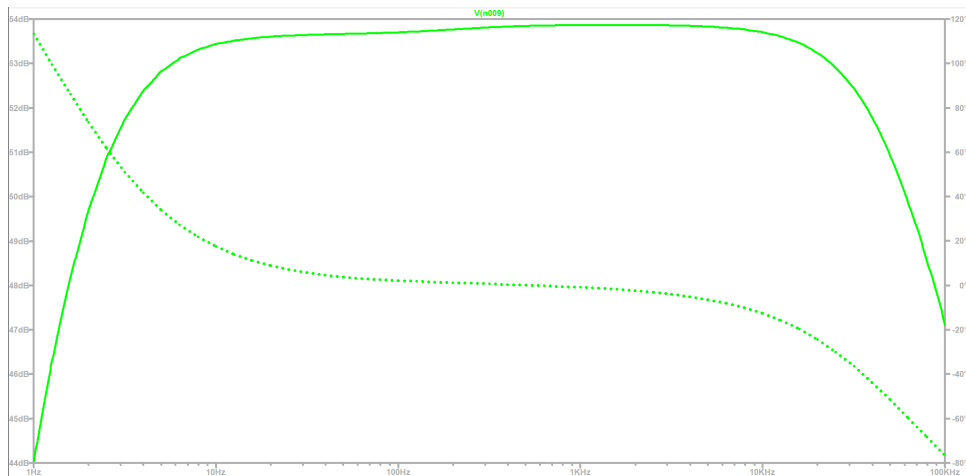


Figure 21: AC analysis — overall gain vs. frequency (LTspice).

Table 5: Overall AC Performance (LTspice)

Metric	Value
Total Mid-band Gain	53.86 dB
System Bandwidth	2.3 Hz - 56.8 kHz
Phase Margin	179.99°

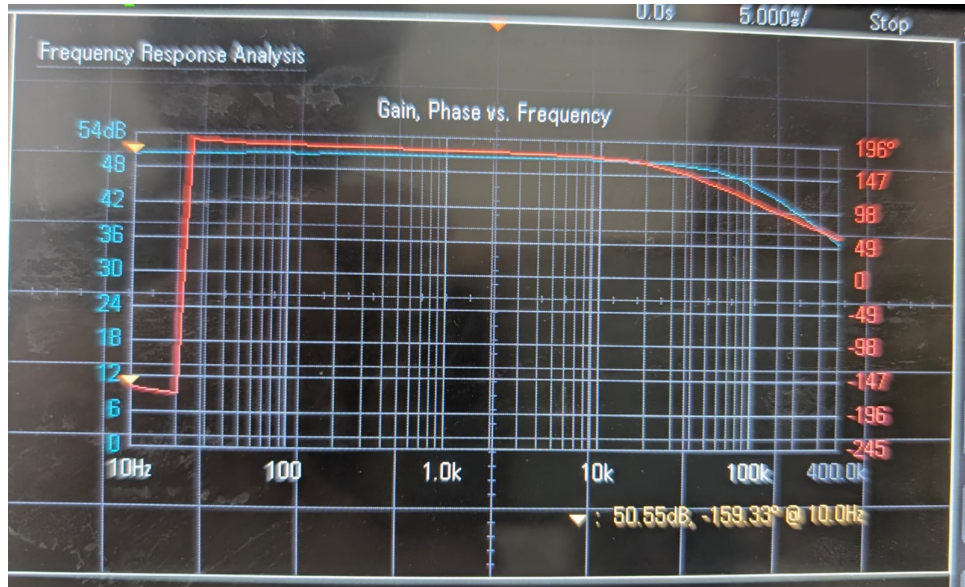


Figure 22: Frequency Analysis of Hardware

6.1.2 Transient Analysis

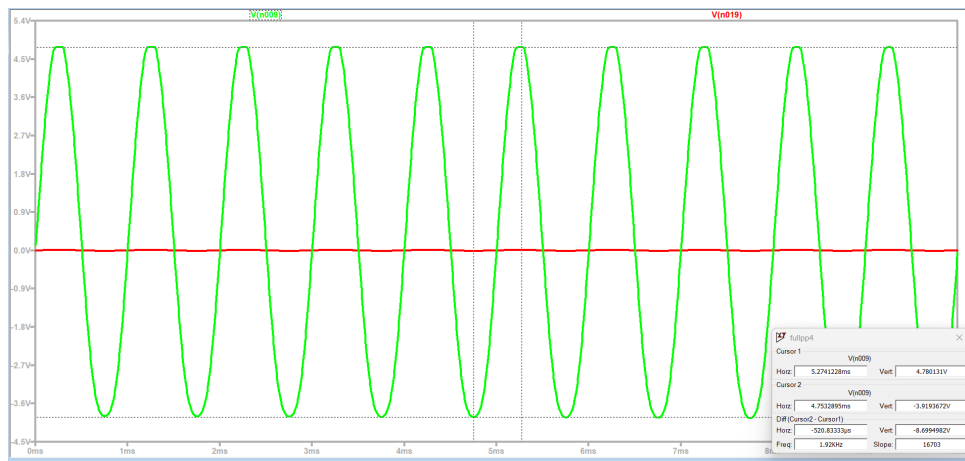


Figure 23: Transient analysis — input vs. output waveform (LTspice and real circuit).

Table 6: Overall Transient Measurements

Metric	Value
Input Peak-to-Peak	20 mV
Output Peak-to-Peak	8.7 V
Voltage Amplification (Linear Range)	435

6.2 Final Gain and THD

The total mid-band gain was measured at 52.04 dB (approx. 400 in absolute terms). THD in hardware was found to be 5.24 % at 1 kHz, which is acceptable given the non-idealities of breadboard construction.

6.3 Slew Rate

The simulated maximum slew rate was 27.44 kV s^{-1} , supporting a distortion-free frequency up to 436.72 kHz for a 10 mV input.

7 Conclusion

The design successfully integrated four distinct stages to drive an audio speaker. The use of voltage regulators (7805/7905) ensured stable $\pm 5 \text{ V}$ rails from a $\pm 12 \text{ V}$ source.

Table 7: Summary of Key Performance Metrics

Parameter	Target	Simulation	Hardware
Mid-band Voltage Gain	>500	435 (54.85 dB)	430
Bandwidth (-3 dB)	20 Hz – 20 kHz	0.8 Hz – 50.3 kHz	0.8 Hz – 50.3 kHz
THD (Hardware)	–	5.65 %	6.67 %
Output Power (Sim)	$>1.5 \text{ W}$	1.05 W	0.98 W

References

- [1] A. Sedra and K. Smith, *Microelectronic Circuits*, 7th ed., Oxford University Press, 2014.
- [2] Analog Devices, *LTspice XVII*, [Online]. Available: <https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html>